Project Proposal: Multi-Agent Storyboarding for Personalized Short-Video Generation

COMP 4221: Intro to NLP

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Problem and motivation

Short video is a dominant format for digital content consumption, yet mainstream recommendation systems rely on a two-stage paradigm: matching users to a fixed pool of pre-produced content. This approach limits personalization, especially when a user's niche interests fall outside the available content pool. The urgency of this limitation is highlighted by the *2026 CCF-Kuaishou Large Model Explorer Fund*, which explicitly calls for research into end-to-end personalized short-video generation agents to overcome fragmented recommendation-generation pipelines.

While recent industrial frameworks have explored "generation-as-recommendation," they often rely on a linear instruction generation process and focus heavily on the downstream video rendering pipeline.

In this project, we investigate the crucial upstream **storyboarding and ideation phase**. We propose a multi-agent framework where specialized agents retrieve external information (news, web images), brainstorm, and explore different narrative tracks to create highly personalized, engaging video storyboards on-the-fly, based directly on user watch histories.

Related work

The most closely related work is the recent paper *Recommendation as Generation: Unifying Personalized Video Generation and Recommendation at Industrial Scale* (Cheng et al., 2026), which introduces Video Generation Agents (VGAs) to unify recommendation and video synthesis. However, their pipeline generates a single linear instruction path directly from semantic IDs. Other relevant areas include LLM-based multi-agent collaboration (e.g., AutoGen, ChatDev) and retrieval-augmented generation (RAG). Our project bridges these domains by applying multi-agent collaborative brainstorming and tool-use specifically to personalized video storyboarding.

Data or tasks

We will evaluate the system using two sets of inputs:

1. **Public Dataset (Automated Eval):** A sampled subset of user profiles from a public recommendation dataset (e.g., KuaiRec or a keyword-adapted MovieLens) to run large-scale automated evaluations.
2. **Real User Histories (Human Eval):** A curated dataset collected from a sample of peers, consisting of their YouTube Shorts watch histories and a brief presentation of their core interests.

The primary task is to generate a comprehensive video storyboard (including scene descriptions, subtitles, and audio cues) tailored to these specific user profiles.

Method and baseline

Method: We will build a multi-agent framework powered by a mix of LLMs (e.g., using smaller models for fast information retrieval and larger models like GPT-4o for complex scriptwriting). These agents will

have access to tools (news databases, image search, historical captions) and will collaboratively debate different storyboard tracks before finalizing a concept. As an extended demo, the pipeline will compile these storyboards into a simple "video sketch" using fetched images, TTS digital voices, and auto-subtitles. **Baseline:** A single-agent, zero-shot pipeline. We will prompt a single frontier LLM (e.g., GPT-4o) with the exact same user history and tool access, asking it to generate the storyboard in one shot. This controlled comparison will isolate and prove the value of our multi-agent brainstorming and routing architecture.

Evaluation

We will employ a mixed-methods evaluation to compare our method against the baseline:

- **Automated Metrics (LLM-as-a-judge):** Using a strong LLM to score the generated storyboards on (a) relevance to the user's historical profile, (b) completeness, and (c) creativity/diversity to allow for large scale grading over big datasets.

- **Human Evaluation (Blind A/B Testing):** The real users from our curated dataset will blindly review the storyboards (or video sketches) generated by both the baseline and our system, voting on which video they would actually prefer to watch in their feed.

Resources and risks

Resources: API credits for LLM providers (e.g., OpenAI API), access to basic search/retrieval endpoints, and local compute for orchestrating the agent framework and rendering the video sketches.

Risks: Multi-agent systems can suffer from long generation latency and compounding API costs. Furthermore, processing extensive watch histories may push against context-window limits, requiring efficient summarization strategies.

Work plan

For the October 29 milestone, our minimum viable pipeline (MVP) consists of some of the data collection from the public dataset, the working single-agent baseline, and a preliminary version of the multi-agent framework. We have divided ownership of the project components as follows:

- **Data & Baseline (Milestone MVP):** Filtering the public dataset, gathering real user histories, and implementing the single-agent baseline to guarantee preliminary results. **[Owner: Antoine]**
- **Multi-Agent Framework (Milestone MVP):** Initial development of the multi-agent storyboarding architecture, prompt engineering, and basic integration of retrieval tools. **[Owner: Charles]**
- **Evaluation & Demo Renderer:** Design of the LLM-as-a-judge rubric, executing automated evaluations on the public dataset, and implementing the final demo sketch renderer (TTS + Images). **[Owner: Max]**

Practical Test

> *On* ***personalized short-video storyboarding****, our change* ***to a tool-augmented multi-agent brainstorming framework*** *should improve* ***content creativity and alignment with niche user interests*** *relative to* ***a single-agent zero-shot generation baseline****, measured by* ***LLM-as-a-judge relevance scores and blind human preference testing*** *under a budget of* ***$50 in API credits and local compute****.*